Date

Azure is highly available cloud environment with uptime guarantees.
SLA's can be between ~~client~~ service provider and customer. eg also be between an organizations in IT dep and business users (**Service level Agreement**)

→ Highly Available services do come at an extra cost. Each Azure service has its own SLA.

→ **Scalability**:. Adjust resources to meet demand. If more traffic and your system are overwhelmed, the ability to add more resources to better handle the increased demand. Are not paying ~~for~~ over for services.

Vertical Scaling:- Increasing or decreasing the capabilities of resources

Horizontal Scaling:. Adding or subtracting the No. of resources.

→ **Reliability**:. Ability of system to recover from failures and continue to function.

Cloud is **Decentralized Design** (something is not controlled by a single central point) ⇒ Resources deployed in regions around the world.

→ **Predictability**:. Move forward with confidence. Performance or. Cost Predictibility.

Performance:. Predicting Resources needed. Autoscaling, Load Balancing, H Availa

Cost:. Predicting or forecasting the cost of the cloud spend. Tools like Total Cost of Ownership (TCO) or Pricing Calculator to get estimate on cloud spend

→ **Benefits of security and Governance in the Cloud**:.

→ When deploying IaaS or SaaS ~~eg~~ → support governance and compliance.

Max Control of Security ⇒ IaaS (Physical resources but Manage operating system and installed software

If Automatic Maintenence → PAAS and SAAS best.

Cloud Providers typically handles ⇒ Distributed Denial of service (DDos) attacks. Establish a good governance footprint easily.

→ **Managability of clouds**:. / Managing Cloud resources.

• ~~B~~ Automatically scale. Deploy resources. Monitor health resources. Automatic alerts based on configured metrics.

1 Region → 3 Datacenter → multiple source of power,
distinct cooling, networking.

Date

Manage Cloud Environment and resources through ① Web Portal ② Command line Interface ③ API's. ④ Powershell.

IaaS ⇒ Most Flexible category. / Largest share consumer.

Cloud Provider Respon ⇒ Hardware, Network Connectivity, physical Security

Client Respon → OS, Configuration, Maintenance, Network configure, database, Storage configuration.

Scenarios:: **Lift and Shift Migration** → moving things from physical to IaaS Structure (Virtual Machines)

Testing and Development ⇒ Create ~~the~~ VM environments for developers. ① that were needed to be replicated. You shut down while complete control.

PaaS ⇒ Middle Ground between PAAS & SAAS. Hardware to Storage, Operating systems. Donot Need to worry about licensing or patching for operating systems and databases. (Pay as You Go)

Scenarios: **Development Framework**:: Framework that developers can build upon to develop or customise cloud-based applications.

Analytics or Business Intelligence:: Analyse findings and predicting outcomes to improve forecasting, product design, business Decisions.

SAAS ⇒ Using / Renting a fully developed Application. **Least Flexible but easiest for Non IT knowledge people.**

- Responsible for Data that you put into the system, devices that you allow to connect to the system, users that have access.

Scenarios:: Email Messaging, Business Productivity applications, A Finance and expense tracking.

PHYSICAL INFRASTRUCTURE:: / Architectural Structure.

Datacenters ⇒ Datacenters aren't directly accessible. Grouped into **Azure Regions or Azure Availability zones** ⇒ acheive resiliency & Reliability.

REGIONS :: Geographical Area on the planet that contains at least one but potentially multiple Datacenters that are nearby with low Latency Network

Availability Zones :: Physically separate Datacenter within Azure Region.

1 or more Datacenter. Isolation Boundary (If 1 goes down other works continually

Date

Availability zones are connected through High speed & private fiber optic networks
→ Want to replicate data incase of failure → creating duplicate Hardware Environments → Done by Availability zones.
→ **Build High Availabity in Application** → Co locating compute, storage, networking and data resources in Availabity zone and replicating in other Availabity zones. → Increase in cost

Availabity zones are for → VM, Manged Disk, Load balancers, SQL Database

Services that support availabity zones fall into 3 categories
① **Zonal Service:** Pin resources to a specific zone (VM's, Disk, IP Add)
② **Zone-Redundant Services:** Platform replicates automatically across zones (zone-Redundant storage, SQL database)
③ **Non-Regional Services:** Services are always available even if a part of the region or zone wide outrages.

Event very large that it impacts Multiple Availabity Zones in a single Region →
→ **Region Pairs** ⇒ Paired with same geography → 300 miles at least.
One Region affected the other can take up the work Ex **West US with East US**
Far enough apart to be isolated from regional Disasters.
Adv: ① 1 region prioritized to make sure at least one is restored as quickly as possible.
② Planned updates. minimize downtown and risk of outage of application.
Sovereign Pairs ⇒ instances isolated from main instance. Used for legal purposes

AZURE MANAGEMENT INFRASTRUCTURE:

Resource & Resource Groups:

Anything you create, deploy → resource. Ex VM, VN, Databases cognitive services
→ **Groupings of resources → Resource Groups** → create resource put into RG.
Single Resource → 1 Resource Groups at a time, Not Nested, can be moved but when moved does not associate with former group.
→ 1 action apply to a resource group that action will apply to all the resources within resource groups. add, delete and all.

Date

Azure Subscriptions:- where You Put Resources. Enrollment. As many subscription ~~att~~ we want. Get the money from Enrollment.
Pay Upfront for the Year.
Subscription level cannot enforce
Set Up Alerts when we are out of budget.
Stop certain resources, reduce size of resources for budget management
Problem Resource Groups ⇒ An application cost from resource group it becomes impossible.
Put **Tags** on every resource. So I am able to see the cost/price of the resources in resource groups.
Bill of Quantity ⇒ what Your Billing of Resources are → calculated by Pricing Calculator
Billing Boundary ⇒ separate billings reports.
Management Groups: include multiple Azure subscriptions.
→ How can our applications deployed into another environment?
Through Devops.
Grouping Mechanism. Where you manage Roles. Assign Roles. So that everything is at the same level/uniform. management group tree can support up to six levels of Depth. Subscriptions inherit those permissions.
If Security is managed then why need management Groups?
Bring in someone to manage. company gives Read only to the People and Full access to the third Party.
Azure Compute Services:-
On demand compute services.
Azure Virtual Machine:-
HyperV software Provides the Virtual Machines.
What are reasons we use VM's?
Determing Size of VM's ⇒ 1 core to 4 Gig Ram | General Purpose
1 core to 8 | Memory Optimise ⇒ Double RAM
A series low end usage.

Date

HPC ⇒ High Performance Compute → Very High Band width between CPU and Memory/RAM.

why webservers need High compute? • JIT Compilation.
Doesnot use High compute → caches the pages. → CPU remains very very low. to cater concurrent requests.

→ D series ?

VM scale set ⇒ Scale the No. of VM's we want. Start when traffic comes and remove when ~~it~~ I don't need them (the VM's)

Veritical Scaling ⇒ increasing or decreasing power, will do the VM reboot.
Horizontal Scaling ⇒ adding or removing more nodes.

Virtual Machine Scale Sets (VMSS) ONLY DEAL with Horizontal Scaling.

Imaging Process where OS/VM's will have all the software configured on all the other VM's created. Automaticaly configured.

→ Spot Instance ⇒ You need compute without SLA. used in mining.
Unused capacity at a discounted rate.

Availability Sets: How to configure zones through VM menu.
How your.

Fault Domain ⇒ Area where Faults can occur without destruction.
3 ~~don~~ Fault Domain per datacenter
→ Spread the VM's in Availability Sets in these domains.
Problem ⇒ Doubles the cost.

Update Domain ⇒ Whether a fault or update happens your applications will never be down.

Must Put them in Availability sets (when VM's are created)
only way of Adding VM's to Availability sets is to add them creating VM's. with. same Resource Groups in Availability sets as well.

Everything should be sitting in the same datacenter when working on Data related Projects. as cost factor can come.

High Availability with a datacenter
Even if datacent go down the application will not go down in ~~zone~~ sets.
Extra cost - incremental cost in Availability ~~sets~~ zones.

Date:

VM's access is given and VM need Networking so in order to provide read access we use Custom roles for this. Custom Role goes to all the subscriptions.

Review Update and Fault Domains.

What is Virtualization?

More efficient use of physical computer Hardware – Foundation of cloud Computing creats **Abstraction Layer** over computer Hardware enabling **division of a** single computer's Hardware components **(Processor, Memory, Storage)** into Multiple Virtual Machine (VMs). Each VM are **Independent** computer Having its **own Operating System**. Virtualization enable **efficient use of physical Hardware** and **Investment**. Drives **clouds computing economics**. Enables Cloud Providers to purchase only the **computing resources they Need when they need** and **scale** those resources **cost effectively**.

- **BENEFITS**: ① **Resource efficiency**: No more idling of resources
② **Easier Management** ③ **Minimal Downtime** ④ **Faster Provisioning**

Solutions ⇒ Hyper-V virtualization solution Focuses on virtual versions of server and desktop computers.

Virtual Machines ⇒ Are Virtual Environments that simulate (replicate) a physical computer in software form ⇒ **comprise** ⇒ files of VM Configuration, storage & Snapshots that preserve its State

Azure Subscriptions: Allow you to logically **organize** your **resource groups** an **facilitate Billing**, also Security Boundary as well.

Using Azure Requires Azure subscription – Authenticated and authorized access to Azure products and services.

– An account can have **Multi-subscription account** – Add subscription Boundaries

① **Billing Boundary**: How an Azure account is billed for using Azure. Can create multiple subscriptions for different types of billing requirement – **Azure generates separate billing Reports and invoices for each subscription so that you can organize and manage costs**

For example one for personal use and another for business usage.

Date

Access Control Boundary:- Applies Access-Management Policies at subscription level • Different departments can have different Azure subscription Policies
↳ Allows to manage and control access to resources that users provision.

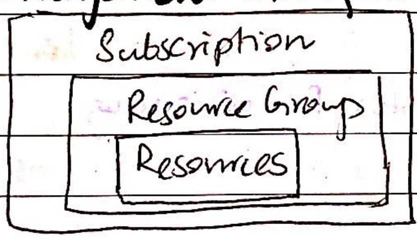
→ Creating Additional Azure Subscriptions:-
Creating Additional Subscriptions of resources or billing management purpose.
We might choose additional Subscription to **separate**.

1- **Environments** ⇒ **Set up** separate environments for **Dev** and **Testing** **security** or to **isolate data** to meet legal requirements.

2- **Organisational Structure** ⇒ To **Reflect** different organizational structures.
Ex limit one team to lower-cost resources while allowing IT Dep for full Range

3- **Billing** ⇒ **For Billing** purposes. cost First aggregated at the subscription level so manage and track costs based on your needs.

→ Azure Management Groups:-



→ **organize Subscriptions into containers** → **Mangement Groups**
↓
Apply Goverance conditions.

All Subscriptions inhert the conditions applied to management Group
Same way **resource group inherits** settings from **subscriptions** and **resource inherit** from **resource groups**.

* **Management Groups give you Enterprise Grade Management at large scale** → Can be **Nested**

→ Management Group, subscriptions and Resource Group Hierarchy:-

• **Create a hierarchy that applies a policy:-** We could limit VM's to be located at a Specific place so it will automatically apply all subscriptions under that Group- **Cannot be changed** by individual resource & Owners.

• **Provide User Access to Multiple Subscriptions:-** If Multiple Subscriptions then group together under 1 Management Group. Then set permissions (RBAC) at Management level → giving user access to all subscription rather than setting permissions for each one separately.

Date

- **10,000** Management groups in 1 single directory.
- Support upto 6 levels of depth → Doesnot Include Root or subscription level
- Each Management Group & subscription can support only **1 Parent**

→ **Virtualization Machines:**

- Provide IaaS in form of a virtualized Server. customize running software

Ideal choice when you need :-

- **Complete control of Operating Systems**
- **Ability to Run custom Software**
- **To use custom hosting Configuration** (Setting up and managing the way website is hosted)

Grouping of VM's - Scale Sets & Availability Sets

→ **Scale Sets** ⇒ Create, manage a group of **identical, load balanced** VM's, As when we create multiple VM's we need to configure identically, then set up network Parameter and monitor utilization which is all done by scale Sets. **Centrally Manage, configure, update VM's.** Automatically increase and decrease done of VM's instances. **Deploy Load Balancer for resources being used efficiently.**

→ **Availability Sets** ⇒ Make systems more reliable and always available. Ensures **VM's stagger updates** (Updates are spread out not happening at the same time) and have **varied Power Network Connectivity** (Don't Rely on same power source

Availability Acheived through VM grouping in 2 ways.

- **Update Domain:-** Grouping of VM's that can be rebooted at the same time. Each Group give 30 minute time to recover.
- **Fault Domain:-** Groups VMS based on their common power source and network switch. By Default VM's are split by Availability sets upto 3 fault Domains. reducing risk of all VM's going down at once.

→ **Usecases of VM's:-**

- **During testing and Development:-** Easy Way to Create OS and delete when not needed
- **When Running Application:-** Substantial economic Benefit. Ex website might need to handle Fluctuations when traffic comes or when resources are not being used. You only pay for what you use.

Date

Azure Virtual Desktop:-

- Another type of Virtual Machine.
- Desktop and application virtualization service that runs on the cloud.
- Use cloud-hosted version of Windows from any location.
- AVD is a service that lets you access a Desktop from any device, no matter what operating system you are using, connect either by special apps or web browsers.

→ **Enhances Security:-**

- provides centralized security management
- uses Microsoft Entra ID (AD) to control who can login and lets you add an extra layer of protection with multiFactor authentication (More than password)
- control what different users can do by assigning roles
- Running in cloud so sensitive information stays in cloud, reducing risk
- Isolated multisessions ensuring privacy and security.

AVD lets you use Windows 10 or 11 entriprise multi-session (only client-based OS that allow concurrent users on single VM.

- More consistent and reliable experience when using diff applications than windows server based system (Windows server).

Azure Containers :-

- Drawback of VM's ⇒ Run only one OS at a time.
- If we want several copies of an app on the same computer without needing multiple VM's, Containers it is.
- Multiple application on single machine while sharing same OS, making them more efficient and flexible than VM's for certain tasks.

Ex: VM is Renting 2 apartments in same building.
Container is renting 2 rooms in same apartment. common resources but each room still serves its own purpose.

→ Virtualized Environment. Multiple VM's on single physical Host, can run multiple containers on single physical Host.

Date

VM ⇒ Have their own operating systems inside them that you need to manage

- Containers ⇒ don't need their own OS - they share the host's OS, making them faster and most efficient to run.

→ containers are lightweight and designed to be created, started, or stopped quickly. They can scale up and down based on demand. If something goes wrong it can be restarted without much hassle.

- Tool for working with containers is Docker and Azure supports it to manage and run containers in the cloud.

→ Diff :- VM's virtualizes Hardware
container virtualizes operating System - Focusing on application rather than hardware or OS.
containerized apps tend to be Smaller in size
Development runtime Environment looks exactly like the production environment
— orchestrated through cluster orchestration → easily deploy and manage containerized application without worrying about which server will host each container.

→ Which to choose?
If complete control then VM's while if portability, performance characteristics and management capabilities containers are good to go.

Azure Container Instances:-
→ fastest and simple way to run a container in Azure without manging vm's
→ Are Platform as a service (PaaS) offering.
→ Allows to upload the container and then the service runs container for you.

Azure Container Apps:-
→ Similar to container Instances
→ Remove management and as PaaS offering.
→ extra benefits → Load Balancing and Scaling.
→ More flexible and efficient without worrying about the technical details.

Date

Azure Kubernetes Service:.

AKS → container orchestration service. An orchestration services manages the lifestyle of containers. If group of containers deploymet then management simpler and more efficient.

Microservice Architecture:. where you break solutions into smaller, independent services. Apps split into halfs like front end hosting, back end hosting etc. → allows separate logical sections that can be maintained, scaled or updated independently. Without impacting other parts.

Azure Functions:.

Event Driven, "serverless" compute option that doesn't require maintaing VM's or containers.

→ service that lets you run small peices of code called functions. only when an event Happens.

"Serverless" → you don't have to manage VM's or containers runing all the time.
↳ Abstraction of servers so you can take mind out of infrastructure concerns

Scenario:.

App build on with VM's or containers, these need to be runing 24/7 for your app to function but using Azure Functions, the code only runs when an event (like trigger) happens. Does not use resources when idle, saving time and money. Runs functions when needed and then goes back to sleep when done.

Benefits of Serverless Compute:.

- No infrastructure management (installing OS)
- Scalability (Application will be working on extreme workloads)
- Only pay for what you use (charge pricing for when code is runing)

Benefits of Azure Functions:.

- Only concerned with the code runing your service
- used when need to perform work in response to an event (REST request
- Scalability based on Demand - when demand is variable.
- Pay for what you use

Functions can be stateless or stateful

Date

- Stateless → By default → behave as if they restart everytime they respond to an event.
- Stateful → Durable functions → context is passed through the function to track prior activity

* Flexibility
If your app's needs to change and you want more control over things like scaling or networking then we can shift to ~~So~~ the functions out of serverless environment and run them in a traditional setup. such as VM's or containers and even isolate the functions.

Application Hosting options:.
- VM's maximum control of hosting environment and allow to configure it exactly how you want.
- most familiar VM's cloud service
- containers → isolate and individually manage different aspects of hosting solutions, could be robust & compelling option.

Hosting options:.

Azure App Services:.
- build and host web Apps, background jobs, backend, RESTful API's in the programing language of your choice without managing infrastructure.
- Automatic scaling and High Availability.
- Supports Windows & Linux.
- Automated deployments from GitHub, DevOps, Git Repo to support Deployment
- Robust Hosting option — use to host apps in Azure
- Focus on building and Maintaining your app only.
- HTTP-based service for hosting web Applications.
- Supports multiple languages .NET, .NET Core, Java, Ruby, Python.

Types of App Services:.
- Web Apps → Host Web Apps by using .NET, .NET Core.
- API apps → Hosting a website. Support, package, Publish API's. can be used by HTTP) client
- WebJobs → Run a program (.exe, Java), script (.cmd) in the same context as webApp
- Mobile Apps → backend for ios and Android.

Date

Virtual Machine VS Container? ::

→ As it is migration (Lift and Shift Migration). Move to cloud in the most Rapid way we opt for lift and shift migration.

⇒ Get the Application containerized and get rid of VM's. don't have to manage OS, manage the environment.

→ Web service, Platform that is well managed, load balancing, integration with DevOps → Web Apps does all of this → due to this Expensive Compute.

Azure ~~M~~ Virtual Networking :—

Backbone of datacenter. When we create a datacenter we start with Networking (cables, routers). Similarly in Azure.

→ Even though we are doing Networking we donot need to understand too much → Area of Software defined Networking

Network Administrators that usually handle Networking.

When we start creating a virtual Network ⇒

10.0.0.0 Address space. /16 ⇒ Subnet mask

4 octs (8 bits). Every octs has 8 bits (0-255). It cannot be signed if it is done signed the value will become half that is to be stored. (divided by 2) 1 bit is used to store the sign.

/24 → 256 Addresses → 3 octs we cannot touch.

In Address space total 32 bits. Subnet Mask says that 24 bits you cannot touch. You can touch the last bits.

Private IP Addresses are defined in a certain Range ⇒ 10, 192 etc

When the subnet mask increases we will get Less IP Addresses and subnet mask less so more IP Addresses to play with.

Max ⇒ /29 subnet mask. → 5 IP Address. At least.

→ Subnet is logical Grouping of IP Addresses in a Network so that we can put certain controls over subnets (subset of Virtual Network

Scenario :: Virtual Network where all the services are running and it is a network where you are logging in. every VM's are there. Virtual load tests, development etc. Provide Access.

Date

Azure Vnets & subnets enable Azure resources such as VM's, web Apps, database to communicate with each other, with users on internet and with client computers.

Azure Network → extension of on-premises network with resources that link with other Azure resources.

→ Azure Network Provide:-

1. Isolating and segmentation
2. Internet communications
3. Communicate between Azure Resources
4. Communicate with on-premises resources
5. Route Network Traffic
6. Filter Network Traffic
7. Connect Virtual Networks.

→ Provides both public & Private End Points → communication between external or internal resources.

Public EndPoints ⇒ Public IP Address. Access from Anywhere in the world.

Private EndPoints ⇒ ~~P~~ Exists within Virtual Network & Private IP from within the address space of that Virtual Network.

1. Isolating and Segmentation:-

- Multiple Isolated Virtual Networks.

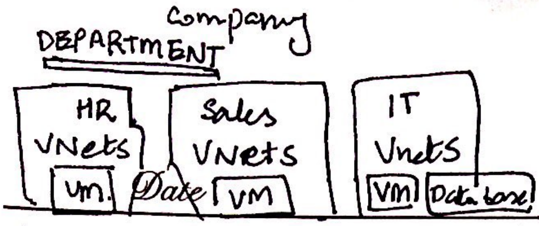
Setup Virtual Network → Define ~~Public~~ Private IP Address Space → Using Public or Private IP Ranges.

IP Range only exist in the Virtual Network. Divide IP address space into "Subnets" and allocate part of Address space to each named subnet.

→ Can configure Virtual Network to either use Internal or external DNS server.

2. Internet Communication:-

Enable Incoming connections from the internet by assigning Public IP Address to Azure resources, or put resource behind a public load balancer.



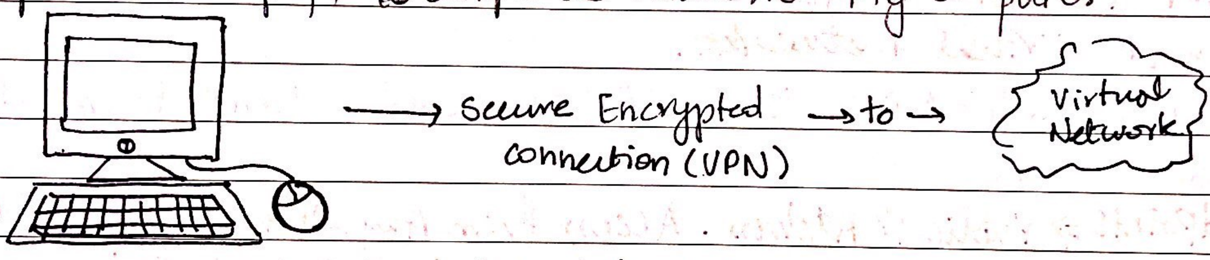
Point to Site ⇒ Single computer to Azure
Site to Site ⇒ Entire office to Azure
ExpressRoute ⇒ Private, High Speed connection to Azure

③ Communicate between Azure Resources:- /Cloud Resources
Azure Resources to communicate securely with each other. 2 ways in doing so:

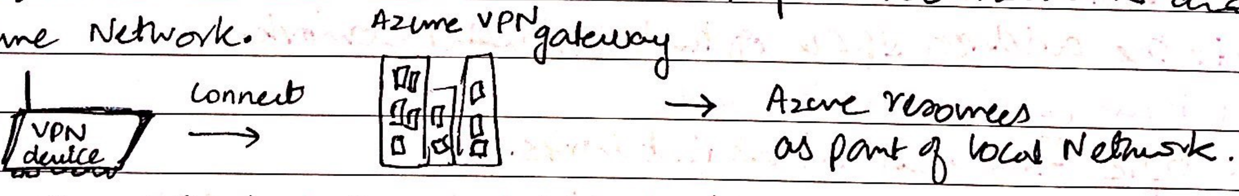
- Virtual Networks can connect not only VM's but other Azure resources
- Service EndPoints allow VNets to connect directly to Azure services like SQL database, storage etc. This connection improves security and ensures that data travels the best possible route between your resources.

④ Communicate between On-premises resources:/ local Resources.
→ Azure VNets allows to link resources together in your on-premises environment and within your Azure subscription. Both local & Cloud create VNets
There are 3 mechanism to acheive this connectivity:-

① **Point-to-Site VPN** (Virtual Private Network):- Connection from a single computer (like laptop) to companies network. My Computer.



② ~~Site to Site~~ VPN :- Connection between on-premises network and then Azure Network.



③ Azure ExpressRoute : Private, dedicated connection to Azure that doesn't use the public internet. Useful for environment for greater bandwidth and even higher levels of security.

⑤ Route Network Traffic:-
Default ⇒ Traffic ⟶ Subnet → on any connected virtual Networks, on premises, internet
Customize ⇒ Route Tables → Define Rules about how traffic should be directed.
create custom tables that control how packets can routed between subnets.
Border Gateway Protocol (BGP):- Share/exchange of routing information between different Networks. Scenario:- Want office network to communicate with Azure, BGP helps share the route so both networks know how to reach each other

Note:: To move a virtual Machine from one VNet to another, you must delete and recreate the Virtual machine in the new VNet.

⊙ Network Peering, connect your private Networks directly together.

Date

⑥ Filter Network Traffic:-

⊙ Network Security Groups (Azure resources) ⇒ Security guards for your network. Set Rules based on source / destination IP, port, protocols.

⊙ Network Virtual Appliances ⇒ Specialized VM's that perform Specific network functions ⇒ Run firewalls, WAN optimising.

⑦ Connect Virtual Networks :-

→ Link VNets together through Network Peering.

→ Peering 2 Vnets to connect directly to each other.

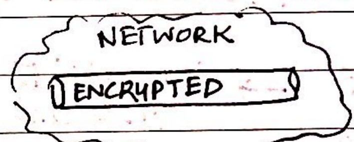
→ Network Traffic in peering Network is "Private". Never entring Public internet

→ Allows Resources in each VNets to communicate.

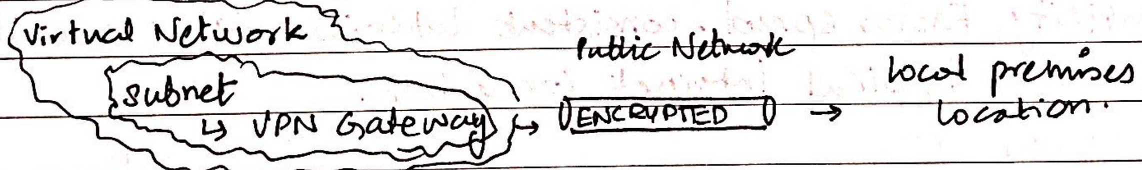
→ VNets can be in Separate Regions → Allows global interconnected Network

User defined Routes (UDR) in Route Tables for controlling them. greater control over network traffic Flow.

Azure Virtual Private Network (VPN):-



→ Connect 2 trusted Private Network to one another over untrusted Network (Public Network).



→ IMP ⇒ Only 1 VPN gateway in each VNet. Use 1 gateway to connect multiple locations e.g Vnets or on-premises data centers.

→ Configure connections:.

⊙ VPN tunnel connection between VPN gateway and other VPN gateway (VNet to Vent)

⊙ Create cross-premises: VPN gateway and on premises VPN device (Site to Site):

VPN Types :-

① Policy-based VPN gateway: Uses Static IP rules to decide which packets to encrypt.

② Route-Based VPN:- Uses dynamic routing to determine the path for encrypted packets, better for changing network setups.

Date

High-Availability Scenarios for VPN Gateways:-

1- Active/Standby:. 2 instances of VPN gateway, one active and one Standby

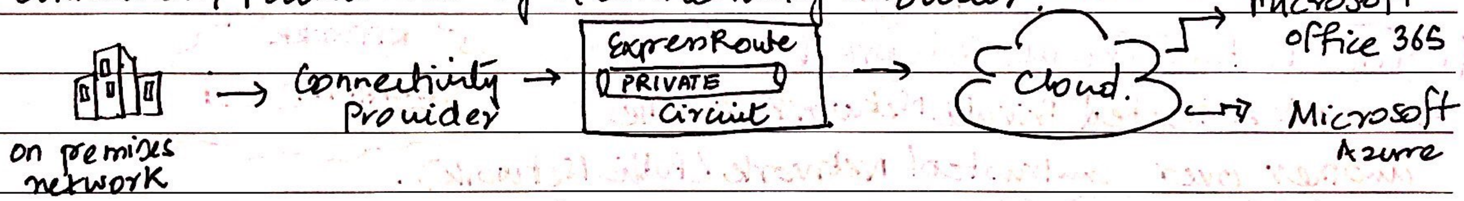
2- Active/Active: Both instances of VPN gateway are active each having its own IP, and separate tunnels

3- ExpressRoute Failover:. Backup connection for ExpressRoute using VPN gateway. If ExpressRoute fails, VPN gateway uses Internet to maintain connectivity.

4. Zone-Redundant Gateways:. Gateways deployed across different availability zones.

Azure ExpressRoute:-

→ Allows to extend on-premises network into Cloud over private connection, facilitated by a connectivity Provider.



→ Reliability, Faster speed, consistent latencies, higher security compared to typical internet connection.

Connectivity Options:- Secure Connection.

- Any to Any (IP VPN) Network
- Point to Point Ethernet Network
- Virtual cross connection through connectivity Provider at a colocation facility

Features & Benefits of ExpressRoute:.

Benefits as connection between Azure & on-premises network:

- Connectivity across all regions in geopolitical Regions
- Global connectivity through Express Route Global Reach
- Dynamic Routing between your network and Microsoft via BGP
- Built in redundancy in every peering location for higher reliability
  ↳ Each connectivity provider uses redundant devices to ensure connection established

Note: Even if ExpressRoute DNS queries, certificate revocation list checking, Azure content Delivery Network Requests sent on the Public Internet.

Date

ExpressRoute enables Direct Access to all the regions:-

- Microsoft office 365
- Microsoft Dynamics 365
- Azure compute services eg VMs
- Azure cloud services e.g Cosmos DB and storage.

Global Connectivity:-

Enable ExpressRoute Global Reach to exchange data across your on premises sites by connecting your expressRoute circuit.

→ Use express Route Global Reach to connect those two facilities, allowing them to connect without transfering data over the public internet.

→ Link express Route circuit creating a private Network bet onprem & ~~cloud~~ network.

Azure DNS:-

Hosting service for DNS domains that provides name resolution by using Microsoft Azure Infrastructure

→ By using this in Azure, you can manage your DNS records using the same credentials, API's, tools, and billing as your Azure services.

Benefits of Azure DNS:-

- Reliability and Performance:
  - ↳ DNS domains are hosted on Azure global network of DNS name servers. uses "Anycast Networking" for fast performance & high Availability
- Security:
  - ↳ Based on Azure Resource Manager. providing Role based Access control, logs, resource locking
- Ease of use:
  - ↳ Azure DNS manage DNS records and DNS for external resources
  - ↳ Integrated with Azure portal, Azure powershell, Azure CLI.
  - ↳ Applications that require automated DNS management uses or integrate with the service by using REST Api and SDK.

Date

- Customizable Virtual Networks:-
  - ↳ Supports private DNS domains.
  - ↳ Allows own custom domain names in your private VNets.
- Alias Records:-
  - ↳ Alias Record Sets. → use to refer an Azure resource such as Azure IP Address Public, Azure Traffic Manager Profile. or Azure Content Delivery Network End Point.
  - ↳ If IP Address of the underlying resource changes, the alias record set seamlessly updates itself during DNS resolution.
  - ↳ The alias record set points to the service instance and the service instance is associated with IP address.

DNS does not allow to buy domain names. You can purchase from. App Service domains or third Party registrars and then manage them.